INCH-POUND

MIL-L-43720C
AMENDMENT 1
19 July 1990

MILITARY SPECIFICATION

LINER, FIELD PACK

This amendment forms a part of MIL-L-43720C, dated 26 July 1988, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

3.3.2: Delete and substitute:

"3.3.2 Heat seal tape. A heat seal tape shall be cut in 1 ± 1/16 inch wide strips of either flat finish semi-clear or flat finish olive green shade tape fabricated in two layers as specified below:

a. Adhesive or bottom layer shall have a minimum thickness of 0.002 inch of low temperature polyurethane adhesive.

b. Barrier or top layer shall have a minimum thickness of 0.004 inch of high temperature polyurethane adhesive.

c. The completed tape shall have minimum thickness of 0.006 inch.

3.3.2.1 Heat sealing. Seams and stitching on outer shell shall be sealed with heat seal tape on the outside of the pack. The entire width of the heat seal tape shall be hot air sealed over the seam or stitching. The tape shall be applied by blowing high temperature air on the thermoplastic adhesive rollers, attaching the tape, and sealing the seam and stitching. The heat sealing machine shall operate on 1200 to 3600 watts with a 115 or 220 volt air heater that has pneumatically activated driven rollers which press and tape onto the seam or stitching being sealed at a maximum rate of 20 feet per minute. All heat seal tape shall be applied without burning, melting, distorting, pinholing, whitening, lifting or stretching and such that a minimum thickness of 3/16 inch overlap is on both sides of the sewn seam. All heat seal tape ends shall overlap a minimum of 3/4 inch beyond the bottom seam or bartack. All tape ends shall be separately sealed on a platen type crossover heat sealer capable of applying pressure and controlling heat. If the heat sealing machine is capable of securing tape ends, a separate heat seal operation is not required. Seam dusting is not applicable to the heat seal tape method. Sealed seams and stitching shall show no leakage when preproduction and end item tested."

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PAGE 5

3.3.5: Delete in entirety.

3.4: Delete in entirety.

3.5 line 1: Delete "figure 1" and substitute "patterns".

PAGE 6

3.7.2, line 4: Delete "sealant or strapping" and substitute "heat seal tape".

Line 6: Delete "(figure 1)"

3.7.2.1: Delete in entirety.

PAGE 7

3.7.3, line 2: Delete "strapped" and substitute "taped".

Line 3: Delete "cemented" and substitute "heat sealed".

Line 8: Delete "adhesive" and substitute "tape".

3.8.2: Delete in entirety.

3.8.3: Delete and substitute:

"3.8.2 Adhesion. The tape shall exhibit bonding strength characteristics as specified in 4.5.4.1."

3.10: Delete and substitute:

"3.10 Patterns. Standard patterns for each of three sizes, which provide allowance of 1/4 inch for all seams, will be furnished by the Government to the contractor for cutting the working patterns. The working patterns shall be duplicates of the Government patterns which show size and notches for proper assembly of all parts. The Government patterns and working patterns shall not be altered in any way."

3.11, line 2: Delete "(see figure 1)"

PAGE 10

4.4.1.1, line 5: Delete "Adhesive and sealing compound" and substitute "Heat seal tape"

Line 6: Delete "Dusting material" and paragraph reference "3.4".

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PAGE 10 (cont'd)

Add "4.4.2.1 Preproduction tests. Prior to heat sealing any end item liners, the basic coated outer shell material shall be evaluated to assure proper heat sealing machine settings, tape alignment, appearance and sealability. An individual examination shall be conducted per 4.5.5.1 through 4.5.5.3 for each seam fix sealing machine at the beginning of each day."

PAGE 11

4.4.3, in Defect column for "Seams", lines 1 and 2: Delete "strapping not securely cemented" and substitute "heat seal tape not securely adhered".

Line 4: Delete "strapping" and substitute "heat seal tape".

Line 5: Delete "Strapping" and substitute "Heat seal tape".

Line 7: Delete "Strapping" and substitute "Heat seal tape".

Line 8: Delete "Strapping" and substitute "Heat seal tape".

Line 10: Delete "strapped" and substitute "taped".

Line 11: Delete "strapping" and substitute "tape".

Line 12: Delete "strapping" and substitute "tape".

Line 13 and 14: Delete and substitute: "Heat seal tape less than 3/16 inch on either side of seam".

Line 15 and 16: Delete and substitute: "Heat seal tape overlapped less than 3/4 inch on bottom of seam".

First "Hem" defect: Delete in entirety

PAGE 12

In Defect column for "Tie cord reinforcing patch", line 1: Delete "cemented" and substitute "seam taped".

In Defect column for "Components and assembly":

Line 9: Delete "strapping" and substitute "tape".

Line 9 and 10: Delete "cemented or cement not evenly".

Line 11 and 12: Delete "or unsealed areas" and substitute ", unsealed areas or failed hydrostatic resistance".

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PAGE 12 (cont'd)

Last Defect on page: Delete in entirety

PAGE 14

Table III: after "Hydrostatic resistance": add "(with flexing)".

Delete "Seam blocking" entry in entirety and substitute "Preproduction testing", "3.3.3" and "4.4.2.1".

Delete "Adhesion of cemented strapping" entry in entirety.

PAGE 16

4.5.3 (a), line 2, after "seam": Delete "," and substitute "and".

Lines 2 and 3: Delete ", and two in the peripheral bottom seam".

Add the following new paragraph:

"(f) Prior to performing the hydrostatic resistance test, the seam shall be flexed ten times in the following manner: The seam shall be grasped using the thumb and fingers of each hand in such a manner that the thumbs are parallel to each other and approximately 1/2 inch beyond (outside) edges of the sealant. With the seam held firmly in the above manner the thumbs shall be brought together so the cloth and the sealant are in contact. The sealed areas shall be flexed vigorously ten time with the cloth and sealant in contact throughout the test. The flexing action shall be such that when fully flexed in either direction there shall be minimum of 2 inches between each thumbnail when the flexed area is fully extended with the test area in contact with itself at all times."

4.5.4 and 4.5.5: Delete:

Add the following new paragraphs:

"4.5.4 Seam seal test.

4.5.4.1 Seam sealing procedure. Cut a 8 X 12 inch (any direction) sample of coated outer shell material down the middle of the 8 inch material width for the entire 12 inch length. Single needle sew the cut edges together using Stitch Type 301 or 401, Seam Type SSa-1, 9-12 inches per inch with thread specified in 3.4. Operate seam-sealing machine according to tape manufacturers recommended setting apply middle of heat sealing tape over face (coated) side of sewn seam leaving at least a 1 inch peal tape of unsealed along the tape ends. There shall be no sign of melting, burning, delamination, puckering, discoloring or any visible sign of degradation of any kind to the coated material. Also,

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the tape shall be flat; with no wrinkles, puckers, or bubbles on its surface. Using the unsealed pull tab section of the tape, slowly pull 3 inches of the sealed tape. The green coated material shall exhibit an even distribution of color change corresponding to the full tape when peeled back.

4.5.4.2 Tape alignment. At random, fold a section of seam sealed material across the width of the tape such that a crease mark is visible when unfolded (press tape on if necessary). Observe that the tape edges remain secure with no sign of delamination at this point. Note that any lifted taped areas, usually denoted by turning opaque, signify either delamination or points in which the heat nozzle of the sealing machine is in misalignment with the tape.

4.5.4.3 Hydrostatic resistance. To determine proper seal, a randomly selected tape section shall withstand the hydrostatic resistance test procedures in 4.5.3, including flexing. As an alternative, a commercially available hydrostatic machine with 4-1/2 inch head may be utilized. The taped specimen shall maintain a hydrostatic pressure at 0.4 psi for 1 minute. In case of dispute, Method 5514 of FED-STD-191 shall be used.

4.5.4.4 Report. If previous taped material observation, tape alignment and hydrostatic resistance results meet the requirements, then record seal machine settings (temperature, pressure and speed) and date on the taped material specimen with indelible ink and continue with daily production utilizing settings. Report all tests and reporting procedures for each day of production. Retain all taped specimens until acceptance of end items has been finalized.

4.5.5 Heat seal tape thickness procedure.

4.5.5.1 Equipment required. Crossover heat sealer, thin glass microscope slides, manila (file folder) cardboard, thickness gauge capable of measuring to the nearest 0.001 inch at 26 pounds psi with 0.25 inch diameter pressure foot.

4.5.5.2 Test method. Cut five 2 inch by 2 inch square manila cardboard specimens and record the thickness in the middle of specimen. Heat the crossover sealer to a temperature of 151 to 154 degrees F (66-68 degrees C). Place a three inch section of seam-seal tape with adhesive side towards the cardboard and tape a cool glass slide on top of the tape. Heat the sandwiched cardboard-tape-glass sample on excessive pressure as to crack the glass slide. Immediately after heat removal, remove the glass slide and peel the tape off cardboard. A uniform adhesive mass will remain attached to the cardboard.

4.5.5.3 Results. Measure thickness of cardboard and adhesive residue at center point of cardboard specimen. The reported thickness of adhesive shall be the difference between the initial center point cardboard thickness and the

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final center point cardboard and adhesive thickness. Membrane or barrier thickness shall be the difference between the initial tape thickness and the reported adhesive thickness. Report thickness to the nearest 0.001 inch."

Figure 1: Delete in entirety.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD
Navy - MC
Air Force - 82
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

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